



Guangdong Technion

Israel Institute of Technology

广东以色列理工学院

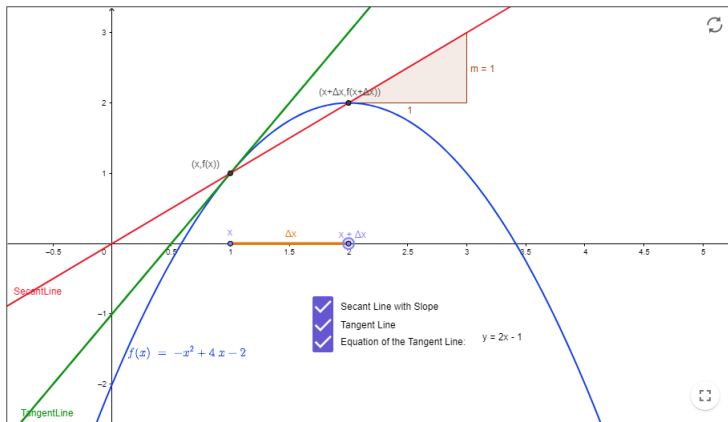
GTIIT

CALCULUS 1 - M
WINTER SEMESTER - 2024

LECTURE 9 - NOVEMBER 11TH



Derivative = limit of the slopes of the secant lines



Derivative = limit of the slopes of the secant lines

Go the Geogebra link to see the animation on [Secant Lines which tend to Tangent Lines](#):

[Secant Lines and Tangent Line](#)

Derivatives: f is differentiable if $f'(x)$ exists

Let $f : (b, c) \rightarrow \mathbb{R}$ be a function. The **derivative** of f at x is

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

This is the **rate of change** of $f(x)$ with respect to x and it is equal to the **slope of the tangent line** at $(x, f(x))$.

Equivalently, by the change of variables $h = x - a$, the **derivative of f at $x = a$** is

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} = \text{slope of the tangent line at } (a, f(a))$$

Higher Order Derivatives

Given a differentiable function f for which also f' is also differentiable, we may consider the **2nd derivative of f** :

$$f''(x) = \frac{d}{dx} f'(x) = \frac{d}{dx} \left(\frac{d}{dx} f \right) (x)$$

- A function can have **higher order derivatives** $f', f'', f''', f^{(4)}$, and so on, each of them exists whenever the previous one is differentiable.

$$\text{Ex: } f(x) = 7x^3 \Rightarrow f'(x) = 7 \cdot 3x^2 \Rightarrow f''(x) = 7 \cdot 3 \cdot 2x \Rightarrow f'''(x) = 42 \Rightarrow f^{(4)}(x) = 0 \checkmark.$$

Then, the **higher order derivatives in this case $f^{(4)}, f^{(5)}$, and $f^{(n)}$ are all zero, for any $n \geq 4$** $\checkmark$.

Higher Order Derivatives

Example: if $f(x) = \sin(x)$ then f has derivatives of any order, but they repeat each 4 steps:

- $f'(x) = \cos(x)$
- $f''(x) = -\sin(x)$
- $f'''(x) = -\cos(x)$
- $f^{(4)}(x) = \sin(x)$, this means that the derivative of order 4 equals f , then this process is cyclic of order 4:

$$f^{(4q+r)} = f^{(r)} : 0 \leq r < 4$$

Higher Order Derivatives

- If $f(x)$ is the position of an object at the time x
- f' is the velocity at the time x , it is the rate of change of the position respect to the time.
- f'' is the acceleration at the time x , it is the rate of change of the slope of the original curve.
- Thus the acceleration function is the derivative of the velocity function and is therefore the **second derivative of the position function**:
Acceleration is the change in velocity (as you feel when speeding up or slowing down in a car.)

Derivatives of the Trigonometric Functions

The following functions are **differentiable** at any $x \in \text{domains}$, such that the quotient is defined:

$f(x)$	$f'(x)$
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$\frac{1}{\cos^2 x}$
$\sec x$	$\frac{\sin x}{\cos^2 x}$
$\csc x$	$\frac{-\cos x}{\sin^2 x}$
$\cotan x$	$\frac{-1}{\sin^2 x}$

Horizontal Tangent Lines

Ex. 1. Find all the values of x such that the graph of f has a horizontal tangent line at $(x, f(x))$:

$$f(x) = e^x \cos x$$

Solution: Horizontal lines have **slope zero**, so

$$0 = f'(x) = e^x [\cos x - \sin x]$$

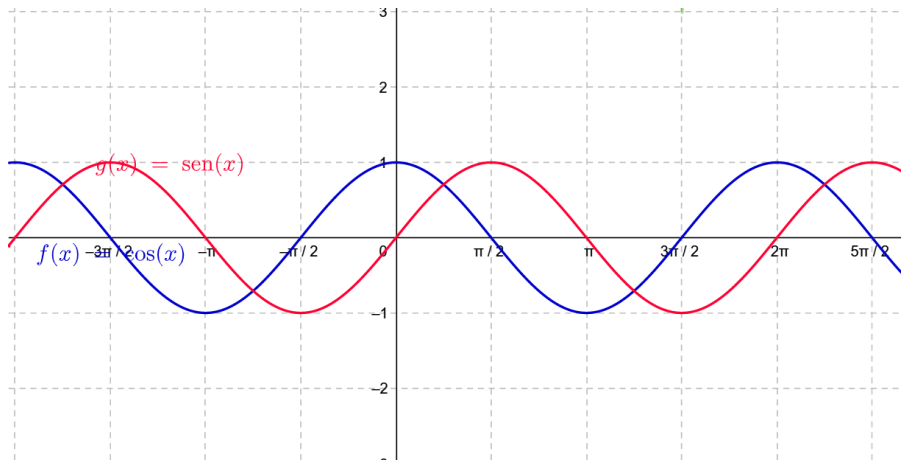
Since $e^x > 0 \forall x \in \mathbb{R}$, then $0 = \cos x - \sin x \Rightarrow \cos x = \sin x$

Hence,

$$x = \frac{\pi}{4} + n\pi : n \in \mathbb{Z}$$



$\sin(x) = \cos(x)$ at any $x = \frac{\pi}{4} + n\pi : n \in \mathbb{Z}$



Theorem: Chain Rule

- If g is differentiable at x and f is differentiable at $g(x)$, then $(f \circ g)(x)$ is differentiable at x and its derivative is given by

$$(f \circ g)'(x) = f'(g(x)) \cdot g'(x)$$

- The composition of differentiable functions is a differentiable function.
- Leibniz Notation: If $y = f(u)$ and $u = g(x)$, then

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$$

Exercise 2.

Compute f' using the rules of the operations, the involved functions and the Chain Rule:

$$a) f(x) = 2^x$$

$$b) f(x) = \log_2(x)$$

$$c) f(x) = 2^x \cdot 4x$$

$$d) f(x) = 5 \ln(3x) - e^x \cdot \sqrt[3]{x^4 + 2} + 1$$

$$e) f(x) = \left[\frac{1}{x^3} - \frac{1}{x} \right] \sin(\pi x)$$

$$f) f(x) = \sqrt{\cos(\sqrt{3x+1})}$$

Solutions.

a) $f(x) = 2^x = e^{\ln 2^x} = e^{x \cdot \ln 2}$ then the derivative requires Chain Rule since the exponent is a product:

$$f'(x) = \frac{d}{dx} [2^x] = \frac{d}{dx} [e^{x \cdot \ln 2}] = [e^{x \cdot \ln 2}] \cdot \ln(2) = 2^x \cdot \ln(2)$$

b) $f(x) = \log_2(x) = \frac{1}{\ln 2} \ln(x)$ then the derivative is

$$f'(x) = \frac{1}{\ln 2} \frac{1}{x}$$

Solutions.

$$\begin{aligned}c) \quad f'(x) &= \frac{d}{dx} [2^x \cdot 4x] = (2^x)' \cdot 4x + 2^x \cdot (4x)' = \ln(2)2^x \cdot 4x + 2^x \cdot 4 \\&= \ln(2)2^x \cdot 4x + 2^{x+2}\end{aligned}$$

$$\begin{aligned}d) \quad f'(x) &= \frac{d}{dx} \left[5 \ln(3x) - e^x \cdot \sqrt[3]{x^4 + 2} + 1 \right] \\&= 5 \cdot \frac{1}{3x} \cdot 3 - e^x \cdot \sqrt[3]{x^4 + 2} - e^x \cdot \frac{1}{3}(x^4 + 2)^{1/3-1} \cdot 4x^3 \\&= \frac{5}{x} - e^x \left[\sqrt[3]{x^4 + 2} + \frac{4x^3}{3(x^4 + 2)^{2/3}} \right]\end{aligned}$$

Solutions.

$$e) \quad f'(x) = \frac{d}{dx} \left[\left(\frac{1}{x^3} - \frac{1}{x} \right) \sin(\pi x) \right]$$

$$\frac{d}{dx} \left[\left(\frac{1}{x^3} - \frac{1}{x} \right) \sin(\pi x) \right] + \left(\frac{1}{x^3} - \frac{1}{x} \right) \frac{d}{dx} [\sin(\pi x)]$$

$$= \left[\left(\frac{-3}{x^4} - \frac{-1}{x^2} \right) \sin(\pi x) \right] + \left[\left(\frac{1}{x^3} - \frac{1}{x} \right) \pi \cos(\pi x) \right] \checkmark$$

$$f) \quad f'(x) = \frac{d}{dx} \left[\sqrt{\cos(\sqrt{3x+1})} \right]$$

$$= \frac{1}{2\sqrt{\cos(\sqrt{3x+1})}} \cdot \left[-\sin(\sqrt{3x+1}) \right] \cdot \frac{3}{2\sqrt{3x+1}}$$

$$= -\frac{3 \sin(\sqrt{3x+1})}{4\sqrt{3x+1} \sqrt{\cos(\sqrt{3x+1})}} \quad \checkmark$$

Exercise 3. Compute the derivatives using Chain Rule:

a) $F(x) = f^n(x)$ for any differentiable function f

b) $G(x) = e^{f(x)}$ for any differentiable function f

c) $H(x) = \ln(f(x))$ for any **positive**, differentiable f

Solution: a) $F(x) = h \circ f$, with $h(x) = x^n \Rightarrow h'(x) = n \cdot x^{n-1}$
then, the derivative of the composition with any
differentiable function f is the following:

$$F'(x) = \frac{d}{dx} [f^n(x)] = n \cdot [f(x)]^{n-1} \cdot f'(x) = n \cdot f^{n-1}(x) \cdot f'(x) \quad \checkmark$$

b) Given $G(x) = e^{f(x)}$ for any differentiable function f , we have $G(x) = g \circ f$, with $g(x) = e^x \Rightarrow g'(x) = e^x$ then, the derivative of the composition with any differentiable function f is the following:

$$G'(x) = \frac{d}{dx} \left[e^{f(x)} \right] = e^{f(x)} \cdot f'(x) \checkmark$$

Logarithmic Derivative

c) Given $H(x) = \ln(f(x))$ for any **positive**, differentiable f , we have

$$H'(x) = \frac{d}{dx} [\ln(f(x))] = \frac{1}{f(x)} \cdot f'(x) \quad \checkmark$$

Exercise 4.

Suppose that $y = -\frac{1}{2}x + 1$ is the equation of the tangent line to the curve $y = g(x)$ at $(2, 0)$ and

$$f(x) = e^{g(x)} + \sqrt{x+7}$$

Find the equation of the tangent line to the curve $y = f(x)$ at $(2, f(2))$.

Solution: Using Chain - Rule to compute the derivative of the composition, we obtain:

$$f'(x) = e^{g(x)}g'(x) + \frac{1}{2\sqrt{x+7}}$$

$$\Rightarrow f'(2) = e^{g(2)}g'(2) + \frac{1}{2\sqrt{9}} = e^0 \left(-\frac{1}{2}\right) + \frac{1}{6} = -\frac{1}{3} \checkmark$$

We know that the equation of the tangent line is

$$L : y = f(a) + f'(a)(x - a)$$

So, in this case, $y = f(x)$ at $(2, f(2))$ we obtain:

$$y = f(2) + f'(2)(x - 2) \Rightarrow L : y = 4 - \frac{1}{3}(x - 2) \checkmark$$

Implicit Differentiation

Some equations do not give expressions of the form
 $y = f(x)$

Some functions, are implicitly defined by a relation between x and y such as

- $x^2 + y^2 = 25$ (1)

or

- $x^3 + y^3 = 6xy$ (2)

In some cases it is **not possible to solve** such an equation in a unique way for y as an explicit function of x .

For instance, if we solve Equation (1), we get:

$$|y| = \sqrt{25 - x^2}$$

⇒ there are **two functions implicitly determined** by (1):

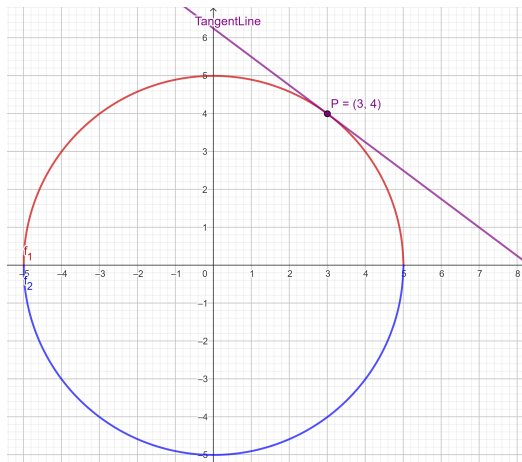
$$f_1(x) = \sqrt{25 - x^2} \text{ and } f_2(x) = -\sqrt{25 - x^2}$$

⇒ The graphs of f_1 and f_2 are the upper and lower semicircles of the circle

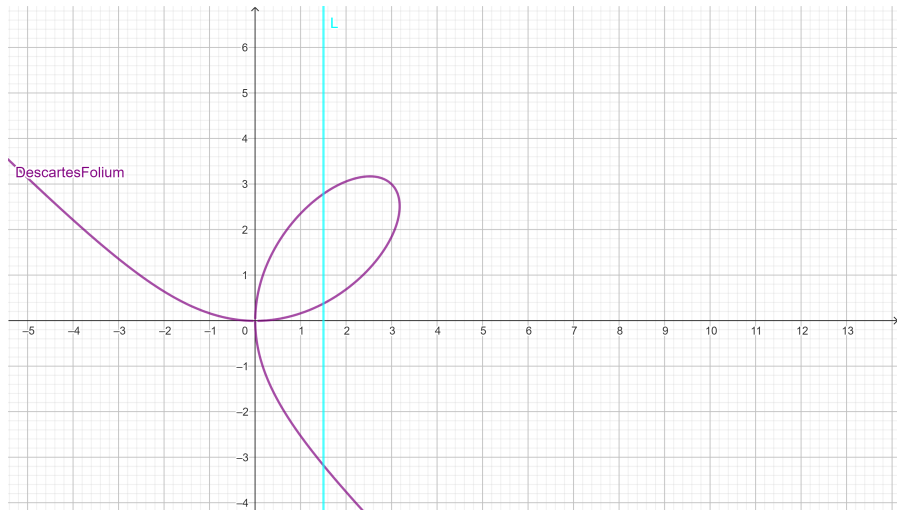
$$x^2 + y^2 = 25$$

Two different solutions of the equation $x^2 + y^2 = 25$

$$f_1(x) = \sqrt{25 - x^2} \text{ and } f_2(x) = -\sqrt{25 - x^2}$$



3 different functions verifying $x^3 + y^3 = 6xy$



- Fortunately, we don't need to solve an equation for y in terms of x in order to find the derivative of $y = y(x)$.
- Instead we can use the method of implicit differentiation:

This consists of **differentiating both sides** of the equation with respect to x and then solving the resulting equation for $\frac{dy}{dx} = y'(x)$.

- The expression obtained for $\frac{dy}{dx}$ is true for any of the different functions which verify the original equation.

Implicit Differentiation

a) If $x^2 + y^2 = 25$, find $\frac{dy}{dx}$.

(b) Find an equation of the tangent to the circle $x^2 + y^2 = 25$ at the point $(3, 4)$.

Solution (a). Differentiate both sides of the equation $x^2 + y^2 = 25$ where $y = y(x)$ using the Chain Rule:

$$\frac{d}{dx} [x^2 + y^2] = \frac{d}{dx} 25 = 0$$

then

$$2x + 2y \cdot \frac{dy}{dx} = 0 \Rightarrow y'(x) = \frac{dy}{dx} = -\frac{x}{y}$$

(b) Find an equation of the tangent to the circle $x^2 + y^2 = 25$ at the point $(3, 4)$.

Replace $(x, y) = (3, 4)$ in the expression:

$$y'(x) = \frac{dy}{dx} = -\frac{x}{y}$$

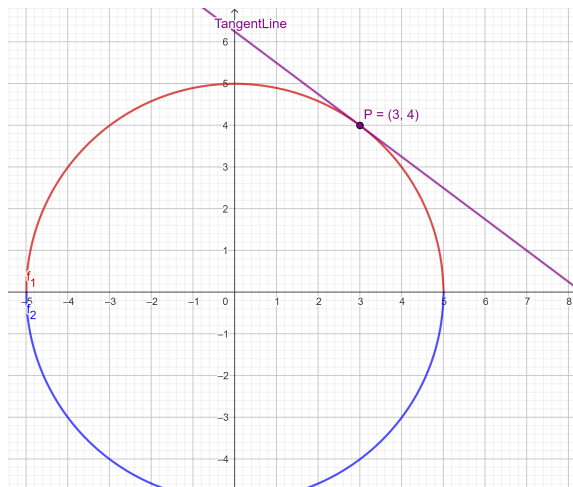
We obtain

$$y'(x) = \frac{dy}{dx} = -\frac{3}{4}$$

Then the equation of the tangent at the point $(3, 4)$ is:

$$y = 4 - \frac{3}{4}(x - 3) \quad \checkmark$$

The equation of the tangent at $(3, 4)$ is: $y = 4 - \frac{3}{4}(x - 3)$



Implicit Differentiation

a) If $x^3 + y^3 = 6xy$, find $y'(x) = \frac{dy}{dx}$.

(b) Find an equation of the tangent to the Descartes Folium $x^3 + y^3 = 6xy$ at the point $(3, 3)$.

c) At what points in the 1st quadrant is the tangent line horizontal?

Implicit Differentiation

a) Find $\frac{dy}{dx}$ if $\sin(x + y) = y^2 \cos x$.

(b) Find an equation of the tangent to the graph of $\sin(x + y) = y^2 \cos x$ at the point $(0, 0)$.

Solution: Derivate both sides:

$$\begin{aligned}\frac{dy}{dx} [\sin(x + y) = y^2 \cos x] \\ = \cos(x + y) \cdot (1 + y') = 2yy' \cdot \cos x - y^2 \sin x \\ \Rightarrow \cos(x + y) + y^2 \sin x = (-\cos(x + y) + 2y \cos x) \cdot y'\end{aligned}$$

Hence

$$\frac{dy}{dx} = y'(x) = \frac{y^2 \sin x + \cos(x + y)}{2y \cos x - \cos(x + y)}$$

From the derivative for any (x, y) in the curve:

$$y'(x) = \frac{y^2 \sin x + \cos(x + y)}{2y \cos x - \cos(x + y)}$$

at $(x, y) = (0, 0)$, we get:

$$y'(0) = -1$$

then the tangent line at $(0, 0)$ is

$$y = -x \checkmark$$

Implicit Differentiation:

$$\sin(x + y) = y^2 \cos x$$



Derivatives of Inverse Functions

Let $f : (a, b) \rightarrow (c, d)$ be a one-to-one differentiable function and let f^{-1} be its inverse, then $f^{-1} : (c, d) \rightarrow (a, b)$ is differentiable and its derivative is

$$(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}$$

Idea of the Proof:

- The composition of f and f^{-1} is the identity function $y = x$:

$$x = f(f^{-1}(x)) \quad \forall x \in (a, b)$$

- Using **implicit differentiation** and **Chain-Rule**, $\forall x \in (a, b)$:

$$1 = \frac{d}{dx} [x] = \frac{d}{dx} [f(f^{-1}(x))] = f'(f^{-1}(x)) \cdot (f^{-1})'(x)$$

Then, we solve

$$(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))} \quad \forall x \in (c, d) \quad \checkmark$$

Example: Suppose that f is a one-to-one differentiable function and $y = 5 - 2x$ is the equation of the tangent line of $y = f(x)$ at $(1, 3)$.

Find the equation of the tangent line of f^{-1} at $(3, 1)$.

Solution: From the hypothesis, we know:

f is one-to-one differentiable $\Rightarrow$ there exists differentiable f^{-1} verifying:

- $f(1) = 3 \Rightarrow f^{-1}(3) = 1$
- $(f^{-1})'(3) = \frac{1}{f'(1)} = -\frac{1}{2}$

So, the equation of the tangent line of f^{-1} at $(3, 1)$ is given by:

$$y = -\frac{1}{2}(x - 3) + 1$$



Example: Find the equation of the tangent line of $\arccos(x)$ at $(0, \frac{\pi}{2})$.

Solution:

- $f(x) = \cos x$ is one-to-one differentiable on $(0, \pi)$
 $\Rightarrow$ there exists the differentiable inverse $\arccos(x)$
- $\cos(\frac{\pi}{2}) = 0 \Rightarrow \cos^{-1}(0) = \frac{\pi}{2}$
- $(\cos^{-1})'(0) = \frac{1}{\cos'(\frac{\pi}{2})} = -\frac{1}{\sin(\frac{\pi}{2})} = -1$

So, the equation of the tangent line of $\cos^{-1}$ at $(0, \frac{\pi}{2})$ is given by:

$$y = \frac{\pi}{2} - x \checkmark$$

Derivatives or the inverse trigonometric f: $\arcsin'(x)$

Write $x = \sin(y) \iff \arcsin(x) = \sin^{-1}(x) = y$, then

$$\Rightarrow 1 = \frac{d}{dx} [\sin(y)] = \cos(y) \cdot \frac{dy}{dx}$$

Equivalently,

$$\frac{dy}{dx} = \frac{1}{\cos(y)}$$

Next, in virtue of the Pythagorean Theorem, for which

$$-\frac{\pi}{2} \leq y \leq \frac{\pi}{2} \Rightarrow \cos y \geq 0 \Rightarrow \cos(y) = \sqrt{1 - \sin^2 y}$$

then

$$\frac{dy}{dx} = \frac{d}{dx}(\arcsin)(x) = \frac{1}{\sqrt{1-x^2}} \checkmark$$

Derivatives of the inverse trigonometric f: $\arctan'(x)$

$x = \tan(y) \iff \tan^{-1}(x) = y$, then

$$\Rightarrow 1 = \frac{d}{dx} [\tan(y)] = \frac{1}{\cos^2(y)} \cdot \frac{dy}{dx}$$

Use the identity

$$1 = \cos^2(y) + \sin^2(y) = \cos^2(y) \left[1 + \frac{\sin^2(y)}{\cos^2(y)} \right] = (1 + \tan^2 y)$$

and solve

$$\frac{dy}{dx} = \cos^2(y) = \frac{1}{1 + \tan^2 y} = \frac{1}{1 + x^2} \checkmark$$

$$\boxed{\frac{dy}{dx} = y' = \arctan'(x) = \frac{1}{1 + x^2} \checkmark}$$

Derivatives of Inverse Trigonometric Functions

are the following:

$$\frac{d}{dx}(\sin^{-1})(x) = \frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}(\csc^{-1})(x) = \frac{-1}{x\sqrt{x^2-1}}$$

$$\frac{d}{dx}(\cos^{-1})(x) = \frac{-1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}(\sec^{-1})(x) = \frac{1}{x\sqrt{x^2-1}}$$

$$\frac{d}{dx}(\tan^{-1})(x) = \frac{1}{1+x^2}$$

$$\frac{d}{dx}(\cot^{-1})(x) = \frac{-1}{1+x^2}$$

Properties of Logarithms

Recall the properties of logarithms with any basis $b > 0, x > 0$:

$$\log_b(x \cdot y) = \log_b(x) + \log_b(y)$$

$$\log_b\left(\frac{x}{y}\right) = \log_b(x) - \log_b(y)$$

$$\log_b(x^a) = a \cdot \log_b(x)$$

Ex:

$$\ln\left[\frac{ab}{c}\right] = \ln a + \ln b - \ln c : \forall a, b, c > 0 \quad \checkmark$$

Logarithmic Differentiation: it provides a method to compute derivatives.

1. Take natural logarithms of both sides of an equation $y = f(x)$ and use the Laws of Logarithms to simplify.
2. Differentiate implicitly with respect to x .
3. Solve the resulting equation for $y'(x)$.

Example of Logarithmic Differentiation: compute the derivative of the given function

$$f(x) = x^{\sqrt{x}}$$

Solution: Take natural logarithms of both sides of $y = x^{\sqrt{x}}$ and then, derivate:

$$\frac{d}{dx} \ln(y) = \frac{1}{y} \cdot y'(x)$$

which is equal to the derivative of right-hand side:

$$\frac{1}{y} \cdot y'(x) = \frac{d}{dx} \left[\ln \left(x^{\sqrt{x}} \right) \right] = \frac{d}{dx} \left[\sqrt{x} \cdot \ln x \right] = \frac{1}{2\sqrt{x}} \cdot \ln x + \sqrt{x} \cdot \frac{1}{x}$$

Then, solve

$$f'(x) = y' = x^{\sqrt{x}} \cdot \left[\frac{1}{2\sqrt{x}} \cdot \ln x + \sqrt{x} \cdot \frac{1}{x} \right] \quad \checkmark$$

Simplify:

$$f'(x) = x^{\sqrt{x}} \cdot \left[\frac{2 + \ln x}{2\sqrt{x}} \right] \checkmark$$

Example of Logarithmic Differentiation: compute the derivative of the given function

$$f(x) = \frac{x^{3/4}\sqrt{x^2+3}}{(5x+2)^7}$$

Solution: 1. Take natural logarithms of both sides of $y = f(x)$ and use the property:

$$\ln \left[\frac{ab}{c} \right] = \ln a + \ln b - \ln c : \forall a, b, c > 0$$

Then

$$\begin{aligned} \ln y &= \ln \left[\frac{x^{3/4}\sqrt{x^2+3}}{(5x+2)^7} \right] = \ln(x^{3/4}) + \ln \sqrt{x^2+3} - \ln((5x+2)^7) \\ &= \frac{3}{4} \ln(x) + \frac{1}{2} \ln(x^2+3) - 7 \ln(5x+2) \end{aligned}$$

2. Derivate both sides, using the rule for the left-hand side:

$$\frac{d}{dx} \ln(y) = \frac{1}{y} \cdot y'(x)$$

which is equal to the right-hand side:

$$= \frac{3}{4x} + \frac{1}{2} \frac{2x}{(x^2 + 3)} - 7 \cdot \frac{1}{5x + 2} \cdot 5$$

$$= \frac{3}{4x} + \frac{x}{(x^2 + 3)} - \frac{35}{5x + 2}$$

3. Finally, solve

$$y'(x) = f'(x) = \left[\frac{x^{3/4} \sqrt{x^2 + 3}}{(5x + 2)^7} \right] \cdot \left(\frac{3}{4x} + \frac{x}{(x^2 + 3)} - \frac{35}{5x + 2} \right) \checkmark$$



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CALCULUS 1 - M

WINTER SEMESTER - 2024

LECTURE 10 - NOVEMBER 13TH



The number e as a limit.

$$e = \lim_{x \rightarrow 0} (1 + x)^{1/x}$$

and in equivalent forms

$$e = \lim_{t \rightarrow +\infty} \left(1 + \frac{1}{t}\right)^t \quad \text{and} \quad e = \lim_{t \rightarrow -\infty} \left(1 + \frac{1}{t}\right)^t$$

This number is approximately equal to

$$e \simeq 2.7182818$$

Proof: **How do we come to this?** Let us consider a table of values and then the **formal proof**.

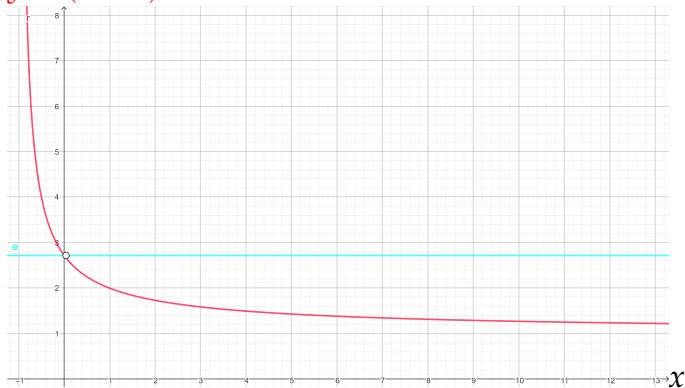
The number e is the result of the following limit

$$\lim_{x \rightarrow 0} (1 + x)^{1/x} = e \simeq 2.7182818...$$

x	$(1 + x)^{1/x}$
10^{-1}	2.5937
10^{-2}	2.7048
10^{-3}	2.7169
10^{-4}	2.718145
10^{-5}	2.718268
10^{-6}	2.718280
$\downarrow$ 0	$\downarrow$ e

$$e = \lim_{x \rightarrow 0} (1 + x)^{\frac{1}{x}} \checkmark$$

$$y = (1 + x)^{\frac{1}{x}}$$



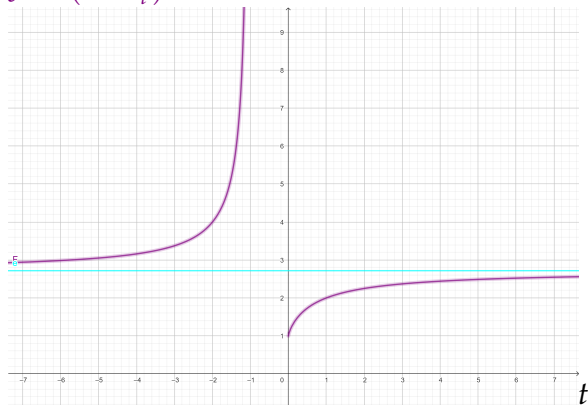
The number e is the result of the following limit

$$\lim_{t \rightarrow \pm\infty} \left(1 + \frac{1}{t}\right)^t = e \simeq 2.7182818...$$

t	$\left(1 + \frac{1}{t}\right)^t$		t	$\left(1 + \frac{1}{t}\right)^t$
10	2.59		-10	2.8679
100	2.70		-100	2.7319
10^3	2.7169		-10^3	2.7196
10^4	2.718145		-10^4	2.7184
10^5	2.718268		-10^5	2.71829
10^6	2.718280		-10^6	2.71828
↓	↓		↓	↓
∞	e		$-\infty$	e

$$e = \lim_{t \rightarrow \pm\infty} \left(1 + \frac{1}{t}\right)^t \checkmark$$

$$y = \left(1 + \frac{1}{t}\right)^t$$



Proof

We know that

$$f(x) = \ln x \Rightarrow f'(x) = \frac{1}{x} \Rightarrow f'(1) = 1$$

then

$$1 = \lim_{x \rightarrow 0} \frac{f(1+x) - f(1)}{x} = \lim_{x \rightarrow 0} \frac{\ln(1+x) - \ln(1)}{x} = \lim_{x \rightarrow 0} \frac{\ln(1+x)}{x}$$

Then, write this derivative by definition:

$$1 = \lim_{x \rightarrow 0} \ln(1+x)^{1/x} \left(* \right)$$

Next, use that the composition of f with its inverse is the identity and both are continuous:

$$f(f^{-1}(x)) = x \Rightarrow e^{\ln x} = x \quad \forall x > 0$$

Apply exponential function to both sides of (*):

$$1 = \lim_{x \rightarrow 0} \ln(1+x)^{1/x} \quad (*) \Rightarrow e = e^1$$

$$= \lim_{x \rightarrow 0} e^{\ln(1+x)^{1/x}} = \lim_{x \rightarrow 0} (1+x)^{1/x}$$

Hence,

$$e = \lim_{x \rightarrow 0} (1+x)^{1/x}$$

Change variables: take $t = 1/x$ then $t \rightarrow \infty$ as $x \rightarrow 0$ so

$$e = \lim_{t \rightarrow \infty} \left(1 + \frac{1}{t}\right)^t \quad \checkmark$$

Example 1: Compute

$$\lim_{x \rightarrow 0} (1 + 2x)^{1/x}$$

Solution: Recall $a^{b \cdot c} = (a^b)^c$, then multiply and divide by 2 in the exponent:

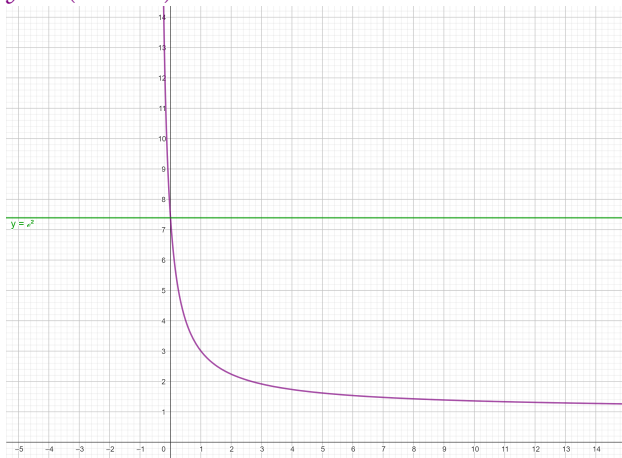
$$L = \lim_{x \rightarrow 0} (1 + 2x)^{2/2x} = \lim_{x \rightarrow 0} \left[(1 + 2x)^{1/2x} \right]^2$$

Next, we use $\lim_{t \rightarrow 0} (1 + t)^{1/t} = e$ for $t = 2x$:

$$L = \lim_{x \rightarrow 0} \left[\underbrace{(1 + 2x)^{1/2x}}_{\rightarrow e} \right]^2 = e^2 \checkmark$$

$$\lim_{x \rightarrow 0} (1 + 2x)^{1/x} = e^2 \checkmark$$

$$y = (1 + 2x)^{1/x}$$



Example 2: Compute

$$\lim_{x \rightarrow \infty} \left[1 + \frac{8}{x}\right]^{x/3}$$

Solution: Recall $a^{b \cdot c} = (a^b)^c$, then multiply and divide by 8 in the exponent:

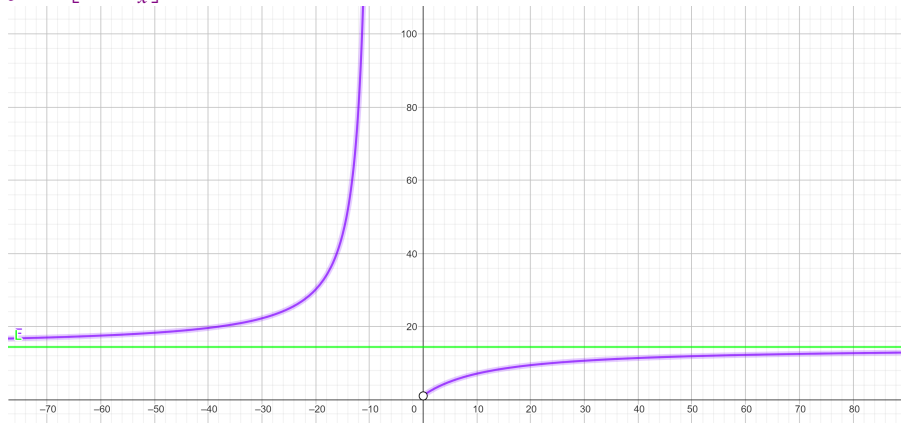
$$L = \lim_{x \rightarrow \infty} \left[1 + \frac{8}{x}\right]^{x/3} = \lim_{x \rightarrow \infty} \left[\left(1 + \frac{8}{x}\right)^{x/8} \right]^{8/3}$$

Next, we use $\lim_{t \rightarrow \infty} \left(1 + \frac{1}{t}\right)^t = e$ for $t = \frac{x}{8}$:

$$L = \lim_{x \rightarrow \infty} \left[\underbrace{\left(1 + \frac{8}{x}\right)^{x/8}}_{\rightarrow e} \right]^{8/3} = e^{8/3} \checkmark$$

$$\lim_{x \rightarrow \infty} \left[1 + \frac{8}{x}\right]^{x/3} = e^{8/3} \checkmark$$

$$y = \left[1 + \frac{8}{x}\right]^{x/3}$$



Example 3: Compute

$$\lim_{x \rightarrow 0} [1 + x^2]^{3/x}$$

Solution: Multiply and divide by x in the exponent:

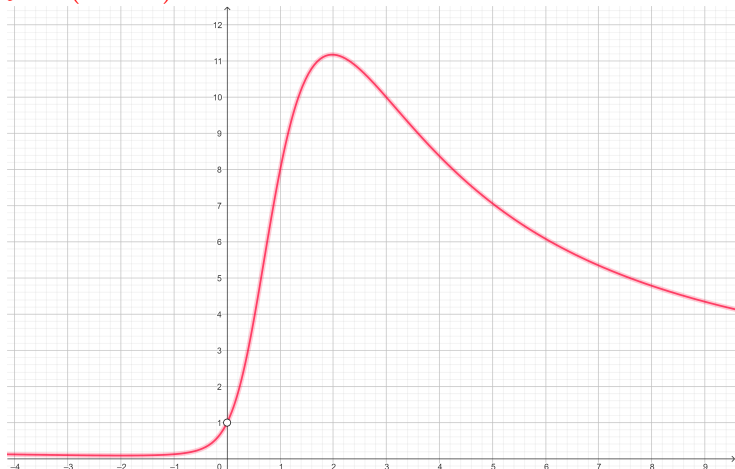
$$L = \lim_{x \rightarrow 0} [1 + x^2]^{3/x} = \lim_{x \rightarrow 0} \left[(1 + x^2)^{1/x^2} \right]^{3x}$$

Next, we use $\lim_{t \rightarrow 0} (1 + t)^{1/t} = e$ for $t = x^2$:

$$L = \lim_{x \rightarrow 0} \left[\underbrace{(1 + x^2)^{1/x^2}}_{\rightarrow e} \right]^{\overbrace{3x}^{\rightarrow 0}} = e^0 = 1 \checkmark$$

$$\lim_{x \rightarrow 0} (1 + x^2)^{\frac{3}{x}} = 1 \quad \checkmark$$

$$y = (1 + x^2)^{\frac{3}{x}}$$



Example 4: Compute

$$\lim_{x \rightarrow \infty} \left[\frac{3^x - 1}{3^x + 1} \right]^{4^x}$$

Solution Call $L = \lim_{x \rightarrow \infty} \left[\frac{3^x - 1}{3^x + 1} \right]^{4^x}$

$$= \lim_{x \rightarrow \infty} \left[1 - \frac{2}{3^x + 1} \right]^{4^x} = \lim_{x \rightarrow \infty} \left[\left(1 - \frac{2}{3^x + 1} \right)^{-\frac{(3^x + 1)/2}{1} \cdot \frac{-2 \cdot 4^x}{3^x + 1}} \right]$$

So, we use the previous limit for the variable

$$t = -\frac{2}{3^x + 1} \rightarrow 0 \text{ as } x \rightarrow \infty:$$

$$\lim_{x \rightarrow \infty} \left[\left(1 - \frac{2}{3^x + 1} \right)^{-(3^x + 1)/2} \right] = \lim_{t \rightarrow 0} (1 + t)^{1/t} = e$$

So, $L = \lim_{x \rightarrow \infty} \left[\underbrace{\left(1 - \frac{2}{3^x + 1}\right)^{-(3^x + 1)/2}}_{\rightarrow e} \right]^{\frac{-2 \cdot 4^x}{3^x + 1}}$ Next,
compute the limit of the exponent:

$$\lim_{x \rightarrow \infty} \frac{-2 \cdot 4^x}{3^x + 1} = \lim_{x \rightarrow \infty} \left(\frac{4}{3}\right)^x \left(\frac{-2}{1 + 1/3^x}\right) = -\infty$$

Hence,

$$L = \lim_{x \rightarrow \infty} \left[\underbrace{\left(1 - \frac{2}{3^x + 1}\right)^{-(3^x + 1)/2}}_{\rightarrow e} \right]^{\overbrace{\frac{-2 \cdot 4^x}{3^x + 1}}^{\rightarrow -\infty}} = 0 \quad \checkmark$$

Hyperbolic Functions.

Combinations of the exponential functions e^x and e^{-x} arise so frequently in mathematics and its applications, that they deserve to be given special names: **hyperbolic functions**.

- In many ways they are analogous to the trigonometric functions
- They have the same relationship to the hyperbola that the trigonometric functions have to the circle.

Hyperbolic Functions: definitions

- Hyperbolic sine: $\sinh(x) = \frac{e^x - e^{-x}}{2}$
- Hyperbolic cosine: $\cosh(x) = \frac{e^x + e^{-x}}{2}$
- Hyperbolic tangent: $\tanh(x) = \frac{\sinh(x)}{\cosh(x)} = \frac{e^x - e^{-x}}{e^x + e^{-x}}$

And their reciprocals:

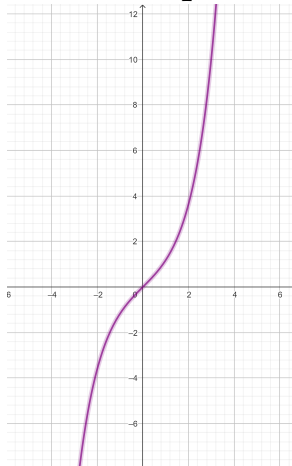
- Hyperbolic cotangent: $\coth(x) = \frac{\cosh(x)}{\sinh(x)} = \frac{e^x + e^{-x}}{e^x - e^{-x}}$
- Hyperbolic cosecant: $\operatorname{csch}(x) = \frac{1}{\sinh(x)} = \frac{2}{e^x - e^{-x}}$
- Hyperbolic secant: $\operatorname{sech}(x) = \frac{1}{\cosh(x)} = \frac{2}{e^x + e^{-x}}$

Hyperbolic Identities

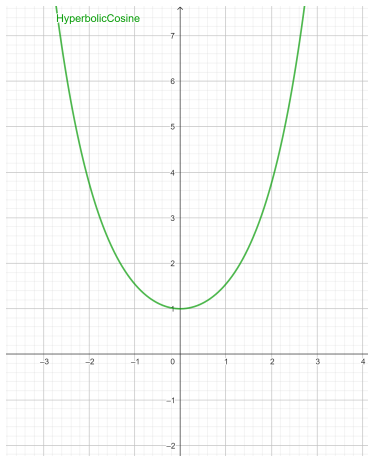
- $\cosh^2(x) - \sinh^2(x) = 1$
- $\sinh'(x) = \cosh(x)$
- $\cosh'(x) = \sinh(x)$
- $\sinh(-x) = -\sinh(x)$
- $\cosh(-x) = \cosh(x)$
- $\sinh(x + y) = \sinh(x)\cosh(y) + \cosh(x)\sinh(y)$
- $\cosh(x + y) = \cosh(x)\cosh(y) + \sinh(x)\sinh(y)$

Hyperbolic Sine and Cosine: graphs

$$\sinh(x) = \frac{e^x - e^{-x}}{2}$$



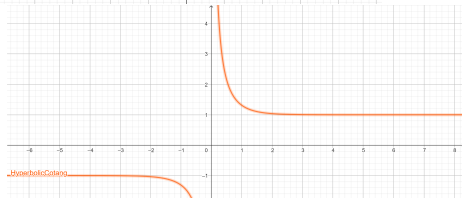
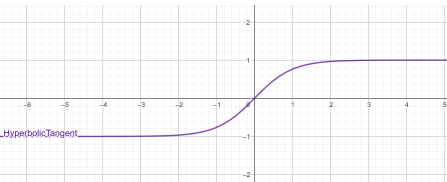
$$\cosh(x) = \frac{e^x + e^{-x}}{2}$$



Hyperbolic tangent and cotangent: graphs

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

$$\coth(x) = \frac{e^x + e^{-x}}{e^x - e^{-x}}$$



Derivatives of the Hyperbolic Functions

The following functions are **differentiable** at any $x \in \text{domains}$, such that the quotient is defined:

$f(x)$	$f'(x)$
$\sinh x$	$\cosh x$
$\cosh x$	$\sinh x$
$\tanh x$	$\frac{1}{\cosh^2 x}$
$\operatorname{sech} x$	$\frac{-\sinh x}{\cosh^2 x}$
$\operatorname{csch} x$	$\frac{-\cosh x}{\sinh^2 x}$
$\operatorname{cotan} x$	$\frac{-1}{\sinh^2 x}$

Inverse Functions of the Hyperbolic Functions

The hyperbolic functions have **differentiable inverse** on certain domains where they are one-to-one; the inverse functions have the following formulas:

- $\sinh^{-1}(x) = \ln \left(x + \sqrt{x^2 + 1} \right) : x \in \mathbb{R}$
- $\cosh^{-1}(x) = \ln \left(x + \sqrt{x^2 - 1} \right) : x \geq 1$
- $\tanh^{-1}(x) = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right) : -1 < x < 1$

Inverse Functions of the Hyperbolic Functions

Proof of $\sinh^{-1}(x) = \ln(x + \sqrt{x^2 + 1}) : x \in \mathbb{R}$:

Let $y = \sinh^{-1}(x) \iff x = \sinh(y)$, then

$$x = \sinh(y) = \frac{1}{2}(e^y - e^{-y}) \Rightarrow 0 = e^y - 2x - e^{-y}$$

So, let us multiply by e^y :

$$0 = e^{2y} - 2xe^y - 1$$

Next, we solve the quadratic equation for $t = e^y$:

$$0 = t^2 - 2xt - 1 \Rightarrow t = e^y = x \pm \sqrt{x^2 + 1}$$

But, since $t = e^y > 0$, only the positive solution is admissible:

$$e^y = x + \sqrt{x^2 + 1}$$

Therefore, the inverse function is

$$\sinh^{-1}(x) = y = \ln(e^y) = \ln\left(x + \sqrt{x^2 + 1}\right) \quad \checkmark$$